

Home Lab Project
Week 2, Day 4

Task: Finalize the cybersecurity home lab architecture, network workflow, data requirements, and technology selections.

Elaboration:

Finalize the overall architecture of the cybersecurity home lab and define how each component will interact within the virtual environment. Document the role of **Oracle VirtualBox, Kali Linux, Metasploitable2, pfSense, the internal LAN, DMZ, and Wireshark.**

Define the **network workflow**, showing how traffic moves through the lab. Establish how **pfSense** controls communication between the network segments, how **Kali Linux** is used for security testing, how **Metasploitable2** functions as the vulnerable target, and how **Wireshark** captures and analyzes the resulting network traffic.

Identify the **data requirements** that will be generated and collected during lab activities, including packet captures, network traffic, scan results, vulnerability findings, firewall events, testing results, and security analysis notes. Determine how this information will be stored and documented for later analysis and reporting.

Finalize the technology choices based on the project's objectives, available resources, security requirements, and learning goals. Confirm that each selected technology has a specific purpose within the lab and contributes to building a controlled network cybersecurity analysis environment.

Expected Outcome:

A finalized lab architecture and technology plan that clearly defines **how the systems connect, how network traffic flows, what security data will be collected, and why each technology is included in the project.**